

Nanyang Technological University
AY2024-25, Semester 1
MH2100 (Calculus III)
Midterm Exam 1

DATE: September 19, 2024

TIME ALLOWED: 90 minutes

FULL NAME _____

MATRICULATION NUMBER _____

TUTORIAL GROUP _____

INSTRUCTIONS TO CANDIDATES

- (1) **DO NOT TURN OVER THE EXAM PAPER UNTIL INSTRUCTED TO DO SO.**
- (2) This midterm exam consists of **ONLY THREE** problems, each consisting of multiple parts.
- (3) Candidates are required to attempt **ALL THREE** problems.
- (4) The total number of points for each problem is indicated at the top right-hand corner of its page.
- (5) Candidates may write anywhere on the blank pages reserved for showing work.
- (6) A **SINGLE** double-sided A4-sized help-sheet with handwritten notes is permitted.

Problem 1**Point allocation:** 15 points

- (a) A curve C in $\mathbb{R}^2$ is parametrized by the vector function $\mathbf{r} : \mathbb{R} \rightarrow \mathbb{R}^2$, defined by

$$\mathbf{r}(t) \stackrel{\text{df}}{=} (t^2 - 1, t^3 - t).$$

[4 points] Find all the unit tangent vectors to C produced by $\mathbf{r}$ at the point $(0, 0)$ on C .

- (b) For each of the following limits, either evaluate it or explain why it does not exist.

(i) [2 points] $\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 y}{x^2 + y^2}.$

(ii) [2 points] $\lim_{(x,y) \rightarrow (0,0)} \frac{xy^3}{x^2 + y^6}.$

(iii) [2 points] $\lim_{(x,y) \rightarrow (1,2)} \frac{\ln(x)}{4x^2 - y^2}.$

- (c) Define a function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ by

$$f(x, y) \stackrel{\text{df}}{=} x^2 + \sin(xy).$$

[5 points] Find all the unit vectors $\mathbf{u}$ in $\mathbb{R}^2$ such that $(D_{\mathbf{u}}(f))(1, 0) = 1$.

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Problem 2**Point allocation:** 15 points

(a) Define a function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ by

$$f(x, y) \stackrel{\text{df}}{=} \begin{cases} \frac{2x^2y + y^3}{x^2 + y^2}, & \text{if } (x, y) \neq (0, 0); \\ 0, & \text{if } (x, y) = (0, 0). \end{cases}$$

- (i) **[3 points]** Determine whether f is continuous at $(0, 0)$. Justify your answer.
- (ii) **[2 points]** Evaluate $f_x(0, 0)$ and $f_y(0, 0)$, if these quantities exist.
- (iii) **[4 points]** Determine whether f is totally differentiable at $(0, 0)$. Justify your answer.

(b) Define a function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ by

$$f(x, y) \stackrel{\text{df}}{=} e^{2x^2 + 3y}.$$

[3 points] Use a linear approximation of f at $(0, 0)$ to estimate the value of $f(0.01, -0.02)$.

(c) Suppose that the functions $f : \mathbb{R}^3 \rightarrow \mathbb{R}$ and $g : \mathbb{R}^3 \rightarrow \mathbb{R}$ satisfy the following properties:

- f is totally differentiable throughout $\mathbb{R}^3$.
- For all $x, y, z \in \mathbb{R}$, we have $g(x, y, z) = f(x - y, y - z, z - x)$.

[3 points] Use the Multivariable Chain Rule to show, for all $x, y, z \in \mathbb{R}$, that

$$g_x(x, y, z) + g_y(x, y, z) + g_z(x, y, z) = 0.$$

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Problem 3**Point allocation:** 20 points

- (a) [4 points] Find a linear equation of the tangent plane to the surface S in $\mathbb{R}^3$, given by

$$xy^2z^3 + x^4y^3z^2 = 4,$$

at the point $(1, -1, 2)$.

- (b) Define a function $F : \mathbb{R}^3 \rightarrow \mathbb{R}$ by

$$F(x, y, z) \stackrel{\text{df}}{=} x^3 + z^2 + ye^{xz} + z \cos(y).$$

Let $S \stackrel{\text{df}}{=} \{(x, y, z) \in \mathbb{R}^3 \mid F(x, y, z) = 0\}$, which is the level surface of F with value 0.

- (i) [2 points] Show that the quantity $F_z(0, 0, 0)$ exists and is non-zero.

Remark (safe to ignore): By the Implicit Function Theorem, we can thus find

- a sufficiently small open disk D in $\mathbb{R}^2$ centered at $(0, 0)$,
- a sufficiently small bounded open interval I in $\mathbb{R}$ centered at 0, and
- a function $z : D \rightarrow I$ that is totally differentiable throughout D ,

such that, within the small region $D \times I$, the surface S is actually the graph surface of z .

- (ii) [4 points] Use the method of implicit differentiation to evaluate $\frac{\partial z}{\partial x}(0, 0)$ and $\frac{\partial z}{\partial y}(0, 0)$.

- (c) Define a function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ by

$$f(x, y) \stackrel{\text{df}}{=} 12x^2 + y^3 - 12xy.$$

- (i) [3 points] Find all the critical points of f .
- (ii) [4 points] Use the Second-Derivative Test to determine the nature of each critical point of f .
- (iii) [3 points] Determine whether f has any absolute extremum. Justify your answer.

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